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29 October 1964
Superseding
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16 July 1957

MILITARY SPECIFICATION

ARMING DEVICE (UNLOADED-LOADED) MARK 2 MODS 1 AND 3

(FOR EXPLODER MECHANISM MARK 19 SERIES)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification states the requirements for the procurement of Arming Device Mk 2 Mods 1 and 3 (Unloaded) and the requirements for the loading of Arming Device Mk 2 Mods 1 and 3. Arming Device Mk 2 Mods 1 and 3 (Unloaded) perform the final electrical switching and mechanical alignment necessary to arm the exploder and, when loaded, contain the primary explosive train which detonates the torpedo warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-350	Jolt Test for Use in the Production of Fuses
MIL-STD-352	Forty-Foot Drop Test for Use in the Production of Fuses

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 275919	Arming Device Mk 2 Mod 1 (Explosive loaded) and all Parts and Subassemblies listed Thereon

FSC 1355

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BUWEPS ID 480525	Arming Device Mk 2 Mod 1 (Unloaded) and all Parts and Subassemblies Listed Thereon
BUWEPS ID 495939	Arming Device Mk 2 Mod 3 (Unloaded) and All Parts and Subassemblies Listed Thereon
BUWEPS ID 495940	Arming Devive Mk 2 Mod 3 (Explosive Loaded) and All Parts and Subassemblies Listed Thereon

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Arming Device Mk 2 Mods 1 and 3 (Unloaded) shall consist of a flanged aluminum housing that contains a contact mount assembly and a detonator shutter. Two recesses shall be provided in the underside of the shutter to receive the detonators, and provision shall be made for crimping the booster cup onto the bottom of the arming device housing. When loaded, the arming device contains two Detonators Mk 57 and a tetryl booster pellet.

The shutter, which contains the electrically initiated detonators, is rotated by Exploder Mechanism Mk 19 from the disarmed (safe) position to the armed position. In the disarmed position, the electrical leads of the detonators are electrically shunted to prevent accidental detonation. The detonators are locked in the safe position by the exploder bore rod, which prevents the shutter from rotating. During the arming cycle, the bore rod is withdrawn and the shutter is rotated approximately 90 degrees clockwise. In the armed position, the detonators are placed between two sets of contact fingers and are in line with blow-through holes in the barrier plate which separates the detonators from the booster charge. The contact fingers connect each detonator with the firing circuit. When the detonators are fired, the bottoms of the two blow-through holes are forced through by the explosion of the detonator, initiating the booster charge, which in turn detonates the warhead of the torpedo.

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3.2 General requirements.

3.2.1 Contractor-furnished data. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.3.4).

3.2.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the arming device shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the arming device shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUWEPS LD's 275919 and 480525 (Mk 2 Mod 1) or LD's 495939 and 495940 (Mk 2 Mod 3), as applicable.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the arming device shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD's 275919 and 480525 (Mk 2 Mod 1) or LD's 495939 and 495940 (Mk 2 Mod 3), as applicable.

3.2.5 Moisture, fungus, and corrosion. The arming device shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341 and in accordance with detail requirements specified in the drawings and specifications listed in BUWEPS LD's 275919 and 480525 (Mk 2 Mod 1) or LD's 495939 and 495940 (Mk 2 Mod 3), as applicable.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BUWEPS LD's 275919 and 480525 (Mk 2 Mod 1) and LD's 495939 and 495940 (Mk 2 Mod 3), as applicable.

3.2.7 Interchangeability. Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revision and correction of the applicable drawings. See 3.2.9.

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3.2.8 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection gages. However, such a procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.9 Conflicting requirements. Conflicting requirements arising between this specification and any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Performance requirements and product characteristics.

3.3.1 This specification has been prepared to cover the performance requirements and product characteristics applicable to the manufacture of the unloaded arming device (3.3.2) and to the explosive loading of the arming device (3.3.3). Those activities that both manufacture and load the arming devices under the terms of one contract may omit 3.3.2.1 and 3.3.3.1.

3.3.2 Performance requirements and product characteristics for the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded). The unloaded arming device shall meet the following performance requirements and product characteristics:

3.3.2.1 Operating torque. The shutter, including dummy detonators, when assembled with contact mount assembly without safety spring and when installed in the housing, shall rotate from the unarmed to the armed stop position and from the armed stop position to unarmed stop position. The torque required to rotate the shutter shall be 4 plus 2 minus 3 inch-pounds. The torque required to rotate the shutters in the segments of the arming device as it enters the armed and unarmed positions shall indicate an abrupt increase of at least $\frac{1}{2}$ inch-pound.

3.3.2.2 Armed position resistance. When dummy detonators which have been internally shorted to show a resistance of less than 0.05 ohm between the prongs are inserted in the shutter and rotated to the armed position, the resistance between the terminal pins of the contact mount assembly shall be not greater than 0.1 ohm.

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3.3.2.3 Unarmed position resistance. When individual leads whose resistances are not more than 0.03 ohm each are attached to the prongs of a dummy detonator having open circuit characteristics and the detonator is inserted in the shutter and rotated to the disarmed position, the resistance between the detonator leads shall be not greater than 0.1 ohm.

3.3.2.4 Contact mount resistance. With detonators removed, the resistance between the two terminals of the contact mount assembly shall be not less than 15.4 ohms nor greater than 17.4 ohms.

3.3.2.5 Dielectric strength. On the Contact Mount Assembly, BUWEPS Drawing 883562 (Mk 2 Mod 1) or 1803626 (Mk 2 Mod 3), as applicable, the insulation resistance between either terminal pin and the rivets of the shunt fingers shall be not less than 10 megohms. When the contact mount assembly and shutter are assembled in the housing, without detonators and without the safety spring, the insulation resistance between each terminal pin and the shutter shaft, and between each terminal pin and the housing, shall be not less than 10 megohms.

3.3.3 Performance requirements and product characteristics for the loading of Arming Device Mk 2 Mods 1 and 3. The loaded arming device shall meet the following performance requirements and product characteristics.

WARNING

The tests and inspections of 3.3.3 are to be performed without the tetryl booster pellet (BUWEPS Drawing 959485). However, the safety precautions outlined in 6.5 shall be adhered to when tests and inspections with live detonators are performed.

3.3.3.1 Prior to installation of live detonators, verify that the requirements specified in 3.3.2.2, 3.3.2.3, and 3.3.2.4 have been met.

3.3.3.2 After installation of detonators and assembly of shutter and contact mount assembly within the housing, and with the safety spring installed, apply a torque of 7 plus or minus $\frac{1}{2}$ inch-pounds to the coupler in a direction to rotate the shutter to the armed position. Movement of the shutter out of the disarmed position shall indicate the assembly is defective.

3.3.3.3 Operating torque. After installation of detonators and assembly of shutter and contact mount assembly within the housing, and with the safety spring rendered ineffective, the shutter shall rotate

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from the unarmed to the armed stop position and from the armed stop position to unarmed stop position. The torque required to rotate the shutter shall be 4 plus 2 minus 3 inch-pounds. The torque required to rotate the shutters in the segments of the arming device as it enters the armed and unarmed positions shall indicate an abrupt increase of at least $\frac{1}{2}$ inch-pound.

3.3.3.4 After installation of detonators and assembly of shutter and contact mount assembly within the housing, and with the safety spring installed, the resistance between the terminal pins of the contact mount assembly shall be not greater than 8.35 ohms when in the armed position.

3.4 Environmental requirements.

3.4.1 This specification has been prepared to cover the environmental requirements applicable to the manufacture of the unloaded arming device (3.4.2) and to the explosive loading of the arming device (3.4.3). Those activities that both manufacture and load the arming devices under the terms of one contract may omit 3.4.2.1, 3.4.2.2, and 3.4.2.3.

3.4.2 Environmental requirements for the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded). The unloaded arming device shall meet the applicable performance requirements and product characteristics of this specification after or while being subjected to environmental conditions defined below, when tested as specified in Section 4.

3.4.2.1 Vibration.

3.4.2.1.1 Vibration; ready nonoperating or ready condition. This requirement shall be met with the arming device (unloaded) completely assembled but with no restraint on the coupler, a spring installed (disarmed position), dummy detonators inserted, and a booster cup attached containing a mass simulating the mass of the tetryl pellet. The arming device, so assembled and less pack or packaging and rigidly self-mounted on a vibration device as specified in 4.4.1.4.1.1, shall withstand the vibrations for the time and at the temperatures specified in 4.4.1.4.1.1. These vibrations may occur in any direction. Thereafter, the arming device shall show no damage, shall not have rotated out of the disarmed position, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

3.4.2.1.2 Vibration; operating condition. This requirement shall be met with the arming device (unloaded) completely assembled but with coupler restrained, spring installed, dummy detonators inserted, booster

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cup attached containing a mass simulating the mass of the tetryl pellet, a simulated bore rod installed to render the spring ineffective, and the device rotated to armed position. The arming device, so assembled and self-mounted on a vibration device as specified in 4.4.1.4.1.2 shall withstand the vibrations as defined in 4.4.1.4.1.2 of one g over the frequency range of 20 to 150 cps. These vibrations may occur in any direction. Monitor the circuit during the test to verify continuity in the armed position. Thereafter, the arming device shall show no damage, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

3.4.2.2 Shock; ready nonoperating or ready condition. This requirement shall be met with the arming device (unloaded) completely assembled, but with dummy detonators inserted, booster cup attached containing a mass simulating the mass of the tetryl pellet, spring installed (disarmed position), and with no restraint on coupler. The arming device, so assembled and self-mounted on a shock testing device as specified in 4.4.1.4.2 shall be subjected to a shock requirement of 150 g's peak magnitude, not less than 8 ms pulse duration, and approximating a one-half sine wave pulse, as specified in 4.4.1.4.2. Thereafter, the arming device shall not have rotated out of disarmed position, shall show no damage, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

3.4.2.3 Temperature; high and low operation. The arming device (unloaded), less packaging and packing and mounted as specified in 4.4.1.4.3, shall be subjected to the high and low temperature conditions specified in 4.4.1.4.3. At the conclusion of the test, and while still at the test temperature, the arming device shall show no corrosion or damage, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

3.4.2.4 Operating life. The arming device (unloaded) shall have an operating life of not less than 100 cycles as defined in 4.4.1.4.4.1. Thereafter, the arming device shall show no damage and shall meet all the unloaded arming device performance requirements and product characteristics of this specification, with the exception of 3.3.2.1 (Operating torque).

3.4.3 Environmental requirements for the loading of Arming Device Mk 2 Mods 1 and 3. The loaded arming device shall meet the applicable performance requirements and product characteristics of this specification after or while being subjected to environmental conditions defined below, when tested as specified in Section 4.

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WARNING

The tests and examinations of 3.4.3.1, 3.4.3.2, and 3.4.3.3 are to be performed without the tetryl booster pellet (BUWEPS Drawing 959485). However, the safety precautions outlined in 6.5 shall be adhered to when tests and inspections with live detonators are performed.

3.4.3.1 Vibration.

3.4.3.1.1 Vibration; ready nonoperating or ready condition. This requirement shall be met with the arming device (loaded) completely assembled as shown in BUWEPS Drawing 1803629 or BUWEPS Drawing 490910, as applicable, but with an attached mass simulating the mass of the tetryl pellet, and with no restraint on the coupler. The arming device, so assembled and less pack or packaging and rigidly self-mounted on a vibration device as specified in 4.4.1.5.1.1 shall withstand the vibrations for the time and at the temperatures specified in 4.4.1.5.1.1. These vibrations may occur in any direction. Thereafter, the arming device shall show no damage, shall not have rotated out of the disarmed position, and shall meet all the loaded arming device performance requirements and product characteristics of this specification.

3.4.3.1.2 Vibration; operating condition. This requirement shall be met with the arming device (loaded) completely assembled as shown in BUWEPS Drawing 1803629 or BUWEPS Drawing 490910, as applicable, but with an attached mass simulating the mass of the tetryl pellet, the coupler restrained, a simulated bore rod installed to render the spring ineffective, and the device rotated to the armed position. The arming device, so assembled and self-mounted on a vibration device as specified in 4.4.1.5.1.2, shall withstand the vibrations as defined in 4.4.1.5.1.2 of one g over the frequency range of 20 to 150 cps. These vibrations may occur in any direction. Monitor the circuit during the test to verify continuity in the armed position. Thereafter, the arming device shall show no damage and shall meet all the loaded arming device performance requirements and product characteristics of this specification.

3.4.3.2 Shock; ready nonoperating or ready condition. This requirement shall be met with the arming device (loaded) completely assembled as shown in BUWEPS Drawing 1803629 or BUWEPS Drawing 490910, as applicable, but with an attached mass simulating the mass of the tetryl pellet and with no restraint on the coupler. The arming device, so assembled and self-mounted on a shock testing device as specified in 4.4.1.5.2, shall be subjected to the shock requirement of 150 g's peak magnitude, not less then 8 ms pulse duration, and approximating a one-half sine wave pulse as specified in 4.4.1.5.2. Thereafter, the arming device shall not have

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rotated out of the disarmed position, shall show no damage, and shall meet all the loaded arming device performance requirements and product characteristics of this specification.

3.4.3.3 Temperature; high and low operation. The arming device (loaded), less packaging and packing and mounted as specified in 4.4.1.5.3, shall be subjected to the high and low temperature conditions specified in 4.4.1.5.3. At the conclusion of the test, and while still at the test temperature, the arming device shall show no corrosion or damage and shall meet all the loaded arming device performance requirements and product characteristics of this specification.

WARNING

The following tests and inspections are to be performed on the completely loaded arming device (live detonators and booster pellet). The safety precautions outlined in 6.5 shall be adhered to when performing these tests and inspections.

3.4.3.4 Rough handling tests. These tests are to be conducted on the completely loaded arming device (live detonators and booster pellet).

3.4.3.4.1 Jolt test. Jolt test the arming device in accordance with MIL-STD-350, except as modified by Section 4.

3.4.3.4.2 Forty-foot drop test. Perform the forty-foot drop test on the arming device in accordance with MIL-STD-352, except as modified by Section 4.

3.5 Cleanup. Prior to and after final assembly, all couponents shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. The arming device, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, riveting, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all applicable inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the arming device shall be classified as follows:

a. Preproduction inspection (paragraph 4.4)
b. Acceptance inspection (paragraph 4.5)
c. Periodic production inspection (paragraph 4.4)

4.3 Sampling.

4.3.1 Preproduction sample. A preproduction sample of five arming devices manufactured in accordance with 3.2.2 shall be subjected to the applicable preproduction tests of 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the arming device by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the applicable acceptance inspections of 4.5. These samples shall be subjected to the applicable tests of 4.4 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue with step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

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b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Preproduction and periodic inspection. The preproduction and periodic production samples, after satisfactorily passing the applicable acceptance inspections of 4.5, shall be subjected to the following tests and examinations. Failure of any sample to satisfy all requirements of this specification shall be considered cause for rejection.

4.4.1 This specification has been prepared to cover the preproduction and periodic inspection applicable to the manufacture of the unloaded arming device (4.4.1.4) and the explosive loading of the arming device (4.4.1.5). Those activities that both manufacture and load the arming devices under the terms of one contract may omit 4.4.1.4.1, 4.4.1.4.2, and 4.4.1.4.3, as verification of these requirements is adequately covered by 4.4.1.5.1, 4.4.1.5.2, and 4.4.1.5.3, respectively.

4.4.1.1 Test conditions. Unless otherwise specified, the arming device shall be subjected to preproduction and periodic production tests under the following conditions:

a. Temperature: Room ambient, plus 65 degrees F to plus 95 degrees F (plus 18.33 degrees C to plus 35 degrees C).

b. Altitude: Normal ground.

c. Vibration: None.

d. Humidity: Room ambient to 95 percent relative (maximum).

4.4.1.2 Test equipment required for the preproduction and periodic production inspection of the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

4.4.1.2.1 Two dummy detonators dimensionally equivalent to Detonators Mk 57 (see BUWEPS Drawings 1261834 and 1409500) internally shorted to show a resistance of less than 0.05 ohm between the prongs.

4.4.1.2.2 A simulated tetryl pellet (inert) having the same outline dimensions and density as the explosive pellet (see BUWEPS Drawing 959485).

4.4.1.2.3 A simulated bore rod of 0.186 plus 0.000 minus 0.002 inch diameter capable of rendering the safety spring ineffective (see BUWEPS Drawing 1472658).

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4.4.1.2.4 Two dummy detonators dimensionally equivalent to Detonators Mk 57 (see BUWEPS Drawings 1261834 and 1409500) having open circuit characteristics.

4.4.1.2.5 A fixture capable of simulating the exploder motor (BUWEPS Drawing 1803454) and the exploder accumulator (BUWEPS Drawing 1461552) for purposes of conducting the operating life test. A power supply (BUWEPS Drawing 1277937 or equivalent) shall be used to provide the arming device circuit with useable firing voltage. Suitable arrangement shall be made for charging the firing condenser (see Figure 1) and discharging it through the shunt resistors while in the armed position, that is, switch #1 of Figure 1 shall be closed after the arming device shutter has rotated to the armed position and shall be open at all other times.

4.4.1.2.6 Power Supply, BUWEPS Drawing 1277937, or equivalent.

4.4.1.2.7 Power supply capable of supplying 115 volts, 400 cps, at 0.5 ampere.

4.4.1.3 Test equipment required for the preproduction and periodic production inspection of the loading of Arming Device Mk 2 Mods 1 and 3.

4.4.1.3.1 A simulated tetryl pellet (inert) having the same outline dimensions and density as the explosive pellet (see BUWEPS Drawing 959485).

4.4.1.3.2 A low energy continuity tester capable of verifying electrical continuity through the shunt resistors and Detonators Mk 57 without igniting the detonators.

4.4.1.3.3 A simulated bore rodof 0.186 plus 0.000 minus 0.002 inch diameter capable of rendering the safety spring ineffective (see BUWEPS Drawing 1472658).

4.4.1.4 Preproduction and periodic production inspection for the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

4.4.1.4.1 Vibration

4.4.1.4.1.1 Vibration; ready nonoperating or ready condition. The arming device, assembled as in BUWEPS Drawing 1803629 or BUWEPS Drawing 490910, as applicable, but containing dummy detonators (4.4.1.2.1) and an inert tetryl pellet (4.4.1.2.2), shall be vibrated according to Schedule I in each of the three mutually perpendicular directions as shown in Figure 2; no restraining torque is to be applied to the coupler during the test. Thereafter, the arming device shall show no damage, shall not have rotated out of the disarmed position, shall meet all the unloaded performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

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SCHEDULE I

Frequency Range	Amplitude	Minimum Test Duration Room Temperature	-20°F	+130°F
10 - 30 cps	0.060 ± 0.006 inch	60 min	15 min	15 min
31 - 60 cps	3g ± 0.3g	60 min	15 min	15 min
61 - 100 cps	3g ± 0.3g	60 min	15 min	15 min

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a three to five minute period, for the duration of the test.

As an alternate method the arming device shall be vibrated at specific frequencies differing by not more than five cps in each of the test axes and covering the entire frequency range and amplitude of Schedule I. The total test time shall be not less than that of Schedule I, and the test times at the specific frequencies shall be equally divided.

4.4.1.4.1.2 Vibration; operating condition. With the arming device assembled as in 4.4.1.4.1.1, render the safety spring ineffective by inserting the simulated bore rod (4.4.1.2.3) in the arming device 5/8 plus 1/32 minus 0 inches from the top mounting surface of the shutter. Rotate the shutter to the armed stop position. (Simulated bore rod may be removed prior to vibration testing.) While applying a restraining torque so that the shutter cannot rotate out of the armed position, vibrate the arming device according to Schedule II in each of the three mutually perpendicular directions as shown in Figure 2. Monitor the circuit during the test to verify continuity through the detonators. Thereafter, the arming device shall show no damage, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

SCHEDULE II

Frequency Range	Amplitude	Minimum Test Duration Room Temperature
20 - 60 cps	1g ± 0.1g	15 min.
61 - 150 cps	1g ± 0.1g	15 min.

The frequency shall be continuously varied between each of the limits given at a uniform rate over a three to five minute period for the duration of the test.

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As an alternate method the arming device shall be vibrated at specific frequencies differing by not more than five cps in each of the test axes and covering the entire frequency range and amplitudes of Schedule II. The total test time shall be not less than that of Schedule II, and the test times at the specific frequencies shall be equally divided.

4.4.1.4.2 Shock; ready nonoperating or ready condition. The arming device, assembled as in 4.4.1.4.1.1, shall be shocked in each of the three mutually perpendicular directions, as shown in Figure 2, to 150 g's plus or minus 15 g's peak magnitude, not less than 8 ms pulse duration, and with a pulse shape approximating a one-half sine wave; no restraining torque is to be applied to the coupler during the test. Thereafter, the arming device shall not have rotated out of the disarmed position, shall show no damage, shall meet all the unloaded performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

4.4.1.4.3 Temperature

4.4.1.4.3.1 Low temperature operation. The arming device, assembled as in 4.4.1.4.1.1, shall be placed in a test chamber whose temperature shall be lowered to minus 40 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 hours plus or minus 5 hours. At the conclusion of the test, and while still at the test temperature, the arming device shall show no damage, shall meet all the unloaded arming device performance requirements and product characteristics of this specification, and shall meet the requirement of 3.3.3.2.

4.4.1.4.3.2 High temperature operation. The arming device, assembled as in 4.4.1.4.1.1, shall be placed in a test chamber whose temperature shall be raised to plus 130 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 hours plus or minus 5 hours. At the conclusion of the test, and while still at the test temperature, the arming device shall show no damage, shall meet all the unloaded performance requirements and product characteristics of this specification and shall meet the requirement of 3.3.3.2.

4.4.1.4.4 Operating life.

4.4.1.4.4.1 Definition of one cycle. One cycle of operation for the arming device (unloaded) shall be defined as the rotation of the shutter from the unarmed stop position to the armed stop position, the discharge of the 4.0 microfarad capacitor of Figure 1 through the shunt resistors in the arming device, and the return of the shutter to the unarmed stop position.

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4.4.1.4.4.2 Operating life test. Assemble an arming device with the dummy detonators of 4.4.1.2.4. Rotate the shutter to the armed stop position. The 4.0 microfarad capacitor of Figure 1 shall be discharged through the shunt resistors while in the armed position. The shutter shall then be rotated to the unarmed position. Subject the arming device to not less than 100 complete cycles as outlined herein. Thereafter, the arming device shall show no damage and shall meet all the unloaded arming device performance requirements and product characteristics of this specification, with the exception of 3.3.2.1, (Operating torque).

4.4.1.5 Preproduction and periodic production inspection for the loading of Arming Device Mk 2 Mods 1 and 3.

WARNING

The tests and inspections of 4.4.1.5.1, 4.4.1.5.2, and 4.4.1.5.3 are to be performed without the tetryl booster pellet (BUWEPS Drawing 959485). However, the safety precautions outlined in 6.5 shall be adhered to when tests with live detonators are performed.

4.4.1.5.1 Vibration

4.4.1.5.1.1 Vibration; ready nonoperating or ready condition. The arming device, assembled as in BUWEPS Drawing 1803629 or BUWEPS Drawing 490910, as applicable, with two Detonators Mk 57 but containing an inert tetryl pellet (4.4.1.3.1), shall be vibrated according to Schedule I in each of the three mutually perpendicular directions, as shown in Figure 1; no restraining torque is to be applied to the coupler during the test. Verify that the arming device does not rotate out of the disarmed position during vibration and that the detonators Mk 57 do not detonate. Thereafter, the arming device shall show no damage and shall meet all the loaded performance requirements and product characteristics of this specification.

4.4.1.5.1.2 Vibration; operating condition.

WARNING

Detonators Mk 57 are initiated by low current. Monitor the circuit with a low energy continuity tester.

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With the arming device assembled as in 4.4.1.5.1.1, render the safety spring ineffective by inserting the simulated bore rod (4.4.1.3.3) in the arming device 5/8 plus 1/32 inch minus 0 inch from the top mounting surface of the shutter. Rotate the shutter to the armed stop position and remove the bore rod. While applying a restraining torque so that the shutter cannot rotate out of the armed position, vibrate the arming device according to Schedule II in each of the three mutually perpendicular directions, as shown in Figure 2. Monitor the circuit during the test to verify continuity through the detonators and through the shunt resistors. Thereafter, the arming device shall show no damage and shall meet all the loaded performance requirements and product characteristics of this specification.

4.4.1.5.2 Shock; ready nonoperating or ready condition. The arming device, assembled as in 4.4.1.5.1.1, shall be shocked in each of the three mutually perpendicular directions, as shown in Figure 2, to 150 g plus or minus 15 g peak magnitude, not less than 8 ms pulse duration, and with a pulse shape approximating a one-half sine wave; no restraining torque is to be applied to the coupler during the test. Verify that the arming device does not rotate out of the disarmed position during shock impulses and that the Detonators Mk 57 do not detonate. Thereafter, the arming device shall show no damage and shall meet all the loaded performance requirements and product characteristics of this specification.

4.4.1.5.3 Temperature

4.4.1.5.3.1 Low temperature operation. The arming device, assembled as in 4.4.1.5.1.1, shall be placed in a test chamber whose temperature shall be lowered to minus 40 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 46 hours plus or minus 5 hours. At the conclusion of the test, and while still at the test temperature, the arming device shall show no corrosion or damage and shall meet all the loaded performance requirements and product characteristics of this specification.

4.4.1.5.3.2 High temperature operation. The arming device, assembled as in 4.4.1.5.1.1, shall be placed in a test chamber whose temperature shall be raised to plus 130 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 hours plus or minus 5 hours. At the conclusion of this test, and while still at the test temperature, the arming device shall show no corrosion or damage and shall meet all the loaded performance requirements and product characteristics of this specification.

WARNING

The following tests and inspections are to be performed on the completely loaded arming device (live detonators and booster pellet).

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4.4.1.5.4 Rough handling tests. The arming device, loaded (with live detonators and booster pellet), shall withstand the rough handling tests specified in 3.4.3.4. These tests shall be conducted only when requested by the Government and will normally be restricted to preliminary lots and specially directed programs.

4.4.1.5.4.1 Jolt test. The arming device, loaded (with live detonators and booster pellet), shall conform to the requirements of MIL-STD-350, except that the criteria by which the samples are judged to have withstood this test are that (1) no explosive elements shall explode and (2) the assembly shall be safe to handle as determined by examination. It is not necessary that the arming device be operable after this test.

4.4.1.5.4.2 Drop test. The arming device, loaded (with live detonators and booster pellet), shall conform to the requirements of MIL-STD-352, except that the arming device shall be dropped once and in a direction with the booster in a down position.

4.5 Acceptance inspection.

4.5.1 This specification has been prepared to cover the acceptance inspection applicable to the manufacture of the unloaded arming device (4.5.2.4) and to the explosive loading of the arming device (4.5.2.5). Those activities that both manufacture and load the arming device under the terms of one contract may omit 4.5.2.4.1 and 4.5.2.5.1 as verification of these inspections is adequately covered by 4.5.2.5.3 and 4.5.2.4.2 through 4.5.2.4.4 respectively.

4.5.2 Each arming device delivered under contract shall be subjected to the following tests and examinations as applicable. Failure to meet any requirements specified herein shall be considered cause for rejection.

4.5.2.1 Test conditions. Unless otherwise specified, the arming device shall be subjected to acceptance tests under the following conditions:

a. Temperature: Room ambient, plus 65 degrees F to plus 95 degrees F (plus 18.33 degrees C to plus 35 degrees C).

b. Altitude: Normal ground.

c. Vibration: None.

d. Humidity: Room ambient to 95 percent relative (maximum).

4.5.2.2 Test equipment required for the acceptance inspection of the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

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4.5.2.2.1 Two dummy detonators dimensionally equivalent to Detonators Mk 57 (see BUWEPS Drawings 1261834 and 1409500) internally shorted to show a resistance of less than 0.05 ohm between the prongs.

4.5.2.2.2 A suitable torque indicating instrument capable of measuring torques from 0 to 10 inch-pounds in increments of $\frac{1}{2}$ inch-pounds or less and with an accuracy of plus or minus 0.05 inch-pound.

4.5.2.2.3 Two dummy detonators having open circuit characteristics.

4.5.2.2.4 An ohmmeter with an integral source of $1\frac{1}{2}$ volts direct current (dc) maximum, having a reading accuracy of plus or minus 2 percent and capable of measuring resistances in the order of 0.05 ohm.

4.5.2.2.5 A megohmmeter with an integral source of 500 volts dc and a midscale reading in the order of 10 megohms.

4.5.2.3 Test equipment required for the acceptance inspection of the loading of Arming Device Mk 2 Mods 1 and 3.

4.5.2.3.1 Two dummy detonators internally shorted to show a resistance of less than 0.05 ohm between the prongs.

4.5.2.3.2 An ohmmeter with an integral source of $1\frac{1}{2}$ volts dc maximum, having a reading accuracy of plus or minus 2 percent and capable of measuring resistances in the order of 0.05 ohm.

4.5.2.3.3 Two dummy detonators having open circuit characteristics.

4.5.2.3.4 A suitable torque indicating instrument capable of measuring torques from 0 to 10 inch-pounds in increments of $\frac{1}{2}$ inch-pounds or less and with an accuracy of plus or minus 0.05 inch-pound.

4.5.2.3.5 A simulated bore rod of 0.186 plus 0.000 minus 0.002 inch diameter capable of rendering the safety spring ineffective (see BUWEPS Drawing 1472658).

4.5.2.3.6 A low energy continuity tester capable of measuring electrical resistance of the arming device firing circuit without igniting the detonators.

4.5.2.4 Acceptance inspection for the manufacture of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

4.5.2.4.1 Operating torque test. Insert two dummy detonators (4.5.2.2.1) in the shutter and assemble the shutter and contact mount in the housing, without the safety spring. Attach the torque indicating instrument (4.5.2.2.2) to the operating shaft of the shutter and apply a torque, first in the arming direction and then in the disarming direction, until the shutter comes to a full stop. The torque required to rotate the shutter shall be as specified in 3.3.2.1.

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4.5.2.4.2 Armed position resistance test. Insert the internally shorted dummy detonators (4.5.2.2.1) in the shutter, assemble the shutter and contact mount in the housing, and rotate the shutter to the armed position. Apply the test probes of the ohmmeter (4.5.2.2.4) to the terminal pins of the contact mount assembly. The resistance between the terminal pins of the contact mount assembly shall be as specified in 3.3.2.2.

4.5.2.4.3 Unarmed position resistance test. Attach a short lead wire whose resistance is less than 0.03 ohm to each prong of a dummy detonator (4.5.2.2.3) and insert the detonator in one of the recesses in the shutter. Assemble the shutter to the contact mount assembly and rotate it to the disarmed position. Apply the test probes of an ohmmeter to the leads from the detonator. Repeat this test with the same dummy detonator inserted in the other recess of the shutter. The resistance between the detonator leads shall be as specified in 3.3.2.3.

4.5.2.4.4 Contact mount resistance. This test is performed on the contact mount assemblies, BUWEPS Drawings 883562 (Mk 2 Mod 1) and 1803626 (Mk 2 Mod 3). Apply the test probes of an ohmmeter to the two terminal pins. The resistance shall be as specified in 3.3.2.4.

4.5.2.4.5 Dielectric test. Assemble the shutter and contact mount assembly in the housing, without detonators and without the safety spring. Measure the insulation resistance between either terminal pin and the rivets of the shunt fingers, either terminal pin and the shutter shaft, and between either terminal pin and the housing. The insulation resistance between meggered points shall be as specified in 3.3.2.5.

4.5.2.5 Acceptance inspection for the loading of Arming Device Mk 2 Mods 1 and 3.

WARNING

The tests and inspections are to be performed without the tetryl booster pellet (BUWEPS Drawing 959485). However, the safety precautions outlined in 6.5 shall be adhered to when tests with live detonators are performed.

4.5.2.5.1 Verify, prior to loading the arming device, that the requirements of 4.5.2.4.2, 4.5.2.4.3, and 4.5.2.4.4 have been met.

4.5.2.5.2 Assemble the shutter and contact mount assembly in the housing, with Mk 57 Detonators and the safety spring in position. With the arming device in the unarmed position and with the torque indicating instrument (4.5.2.3.4) attached to the operating shaft of the shutter, apply a torque of $7\frac{1}{2}$ inch-pounds in a clockwise direction. Movement of the shutter out of the disarmed position shall indicate the assembly is defective.

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4.5.2.5.3 Operating torque test. With the arming device assembled as in 4.5.2.5.2 and with the simulated bore rod (4.5.2.3.5) inserted in the arming device 5/8 plus 1/32 minus 0 inch from the top mounting surface of the shutter, attach the torque indicating instrument (4.5.2.3.4) to the operating shaft of the shutter and apply a torque, first in the arming direction and then in the disarming direction, until the shutter comes to a full stop. The torque required to rotate the shutter shall be as specified in 3.3.3.3.

WARNING

Detonators Mk 57 are initiated by low current. Monitor the circuit with a low energy continuity tester.

With the arming device assembled as in 4.5.2.5.2 and in the armed position, apply the test probes of the low energy squib tester (4.5.2.3.6) to the terminal pins of the contact mount assembly. The resistance between the terminal pins of the contact mount assembly shall be as specified in 3.3.3.4.

4.5.2.6 Preservation and packaging. The packaged arming device shall be examined to ensure that it complies fully with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

This specification has been prepared to cover the preparation for delivery applicable to the manufacture of the unloaded arming device (5.1 and 5.2) and to the explosive loading of the arming device (5.3 and 5.4). Those activities that both manufacture and load the arming devices under the terms of one contract may omit 5.1 and 5.2 as verification of these requirements is adequately covered by 5.3 and 5.4 respectively.

5.1 Preservation and packaging of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

5.1.1 Level A. Not applicable. (See 6.2)

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning. Prior to shipping, the arming device shall be thoroughly cleaned as specified in 3.5.

5.1.3.2 Unit packaging. Unless otherwise specified, the packaging of the arming device (unloaded) shall ensure adequate protection against vibration, shock, and temperature and humidity extremes during shipment from the supply source to the loading activity.

5.2 Packing of Arming Device Mk 2 Mods 1 and 3 (Unloaded).

5.2.1 Level A. Not applicable. (See 6.2)

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5.2.2 Level B. Not applicable.

5.2.3 Level C. Unless otherwise specified, the arming device (unloaded) shall be packed to affort protection against damage during direct shipment from the supply source to the loading activity. Containers shall comply with the rules and regulations of the common carrier.

5.3 Preservation and packaging of Arming Device Mk 2 Mods 1 and 3 (Loaded).

5.3.1 Level A.

5.3.1.1 Cleaning. Unless otherwise specified, the arming device (loaded) shall be cleaned in accordance with Method C-1 of Specification MIL-P-116.

5.3.1.2 Drying. Unless otherwise specified, the arming device (loaded) shall be dried in accordance with one or more of the procedures given in Specification MIL-P-116, provided the procedure is not injurious to the item.

5.3.1.3 Preservative application. None.

5.3.1.4 Unit packaging. Each arming device (loaded) shall be packaged in an individual metallic container in accordance with the drawings and other requirements listed in BUWEPS LD 495986 or LD 495987, as applicable.

5.3.2 Level B. Not applicable.

5.3.3 Level C. Not applicable.

5.4 Packing of Arming Device Mk 2 Mods 1 and 3 (Loaded).

5.4.1 Level A. The arming device (loaded), packaged as specified in 5.3.1.4, shall be packed in a wooden box, in quantities of 12 or 24 to a box, in accordance with the drawings and other requirements listed in BUWEPS LD's 495984, 495985, 495988, or 495989, as applicable.

5.4.1.2 Level B. Not applicable.

5.4.1.3 Level C. Not applicable.

5.5 Labeling and marking. Unless otherwise specified, labeling and marking of all shipping containers shall be in accordance with MIL-STD-129. Each container shall be marked with the serial number of the unit it contains.

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6. NOTES

6.1 Intended use. Arming Device Mk 2 Mods 1 and 3 is intended for use in Exploder Mechanism Mk 19 series.

6.2 Packaging and packing. Ordinarily, the unloaded arming device is only transported from the supply source to the Naval Activity where it is loaded and undergoes final assembly. The loaded arming device is then packaged in a specially designed, hermetically sealed metallic container in accordance with the applicable BUWEPS drawings. When it is definitely known at the time of procurement that the unloaded arming device will be stored for an indeterminate period prior to its loading, then the unloaded arming device shall be packaged and packed as specified by the procuring activity.

6.3 Ordering data. The contract or order should specify the following:

6.3.1 Title, number, and date of this specification and any exceptions to this specification, applicable drawings, or other documents.

6.3.2 Packaging and packing requirements if storage is contemplated prior to loading the arming device. See 6.2.

6.3.3 All additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3.4 Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Bureau of Naval Weapons.

6.4 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B. This level shall be used only when it is definitely known that the item will be held in covered storage, either in domestic or overseas locations, for a period of six months or less.

Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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6.5 General safety precautions. The loading, assembly and handling of the item covered by this specification, and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the supplier or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a supplier or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail technical requirements of the specification.

6.5.1 All loading operations should be conducted in a neat and orderly manner.

6.5.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which effect functioning, should insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.5.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

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6.5.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.5.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each

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unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.5.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives and explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

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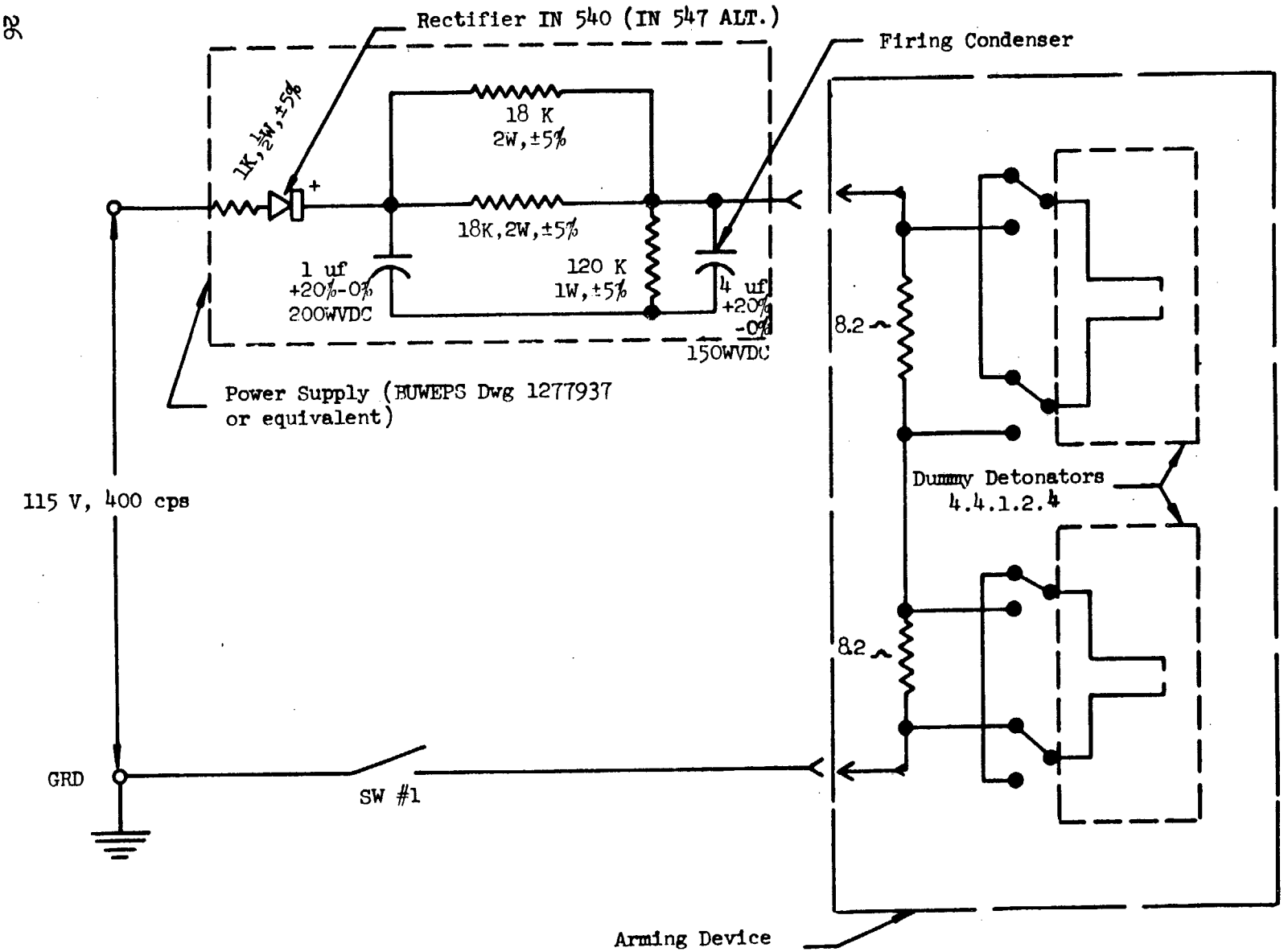
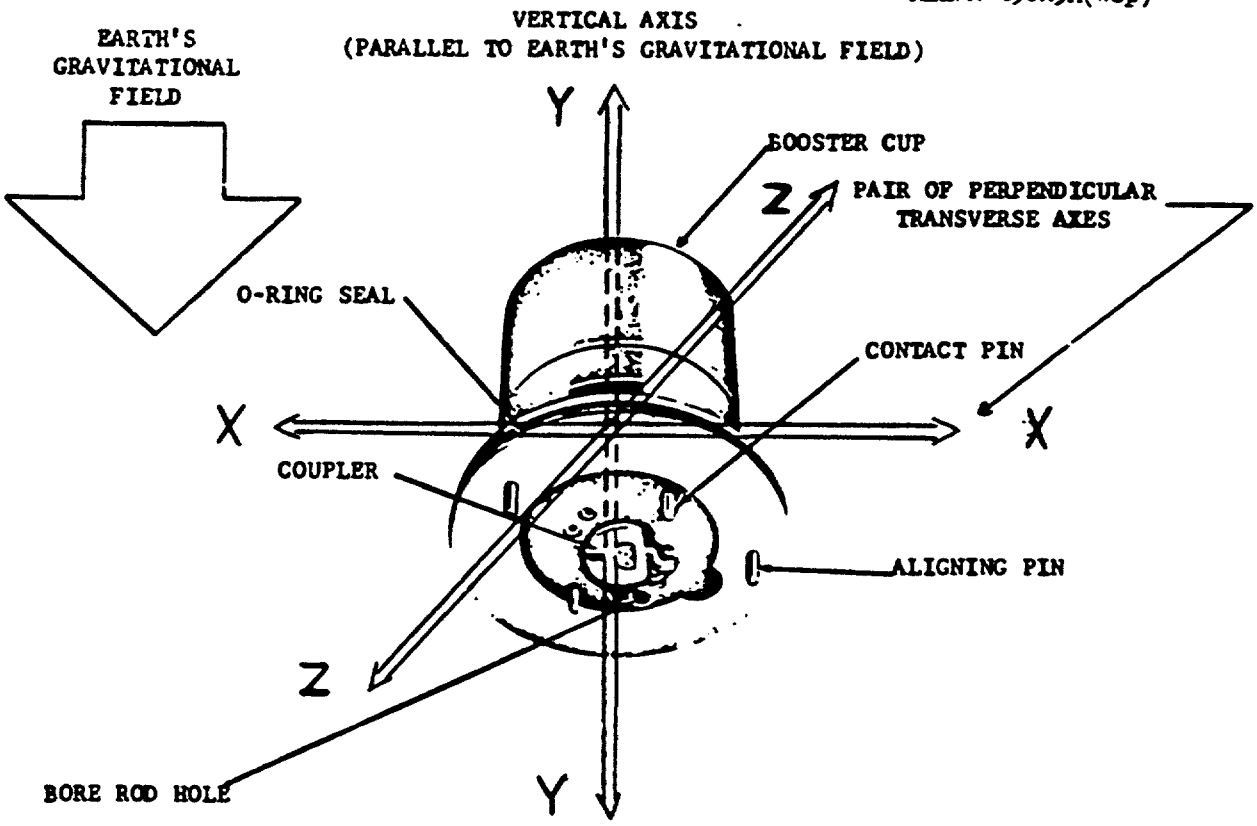


Figure 1. Firing Circuit Schematic
(for test of 4.4.1.4.4)

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NOTES:

1. THE TEST AXES OF THE ARMING DEVICE FOR VIBRATION AND SHOCK TESTS ARE THE VERTICAL AXIS AND ANY PAIR OF PERPENDICULAR TRANSVERSE AXES.

2. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS. (THE ARMING DEVICE COUPLER IS TO BE FACING DOWN FOR ALL TEST DIRECTIONS.)

3. FOR SHOCK, ORIENTATION OF THE TEST ITEM CAN BE CHANGED AS NEEDED TO ACCOMPLISH ALL DIRECTIONS OF TEST.

4. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM SHOCK OR VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

Figure 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-19615A(Wep) ARMING DEVICE (UNLAODED-LOADED) MK 2 MODS 1 & 3 (For Exploder Mechanism MK 19 Series)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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